MILITARY SPECIFICATION

SULFUR: COMMERCIAL GRADE (FOR USE WITH AMMUNITION)

1. SCOPE

1.1 Scope.-Commercial grade sulfur covered by this specification is for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids form a part of this purchase description.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Chemical requirements.-The sulfur shall conform to the requirements on Table I.

TABLE I
Chemical Requirements

	Minimum Percent	Maximum Percent
Sulfur (dry basis)	99.5	------
Moisture	-----	0.5
Ash	-----	0.1
Organic impurities	-----	0.1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of ths inspection requirements as specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material in controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specification or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government. shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

4.2.2 Sampling

4.2.2.1 Lumps.-A representative sample of approximately 2 1/2 pounds shall be taken from each estimated tenth of the lot. The ratio of lumps to fines shall be approximately the same in the sample as in the lot. The 10 samples shall be combined to form a composite sample and handled in the following manner: crush the lumps to reduce the composite sample to laboratory size by the usual quartering process. Place the sample in a clean, dry, glass bottle and stopper tightly. Label the bottle in a manner that will identify the sample with the lot represented. If the sample fails to comply with paragraph 3, the lot shall be rejected.

4.2.2.2 Inspection.-All containers shall be carefully examined and any which show evidence of leakage, loss of contents, objectionable dents, or corrosion, shall be considered as not meeting the requirements of this specification.

4.3 Test methods and procedures.

4.3.1 Moisture, Major defect, Code No. 01001 (see 6.2).-Weigh approximately 50 grams of the sample on a tared watch glass, heat for 12 hours at approximately 82 degrees C., cool in a desiccator and weigh. Calculate the percent of moisture as follows:

$$\text{Percent moisture} = \frac{\text{loss in weight} \times 100}{\text{weight of sample}}$$

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4.3.2 Organic impurities and ash, Major defect, Code No. 02001.-Weigh approximately 15 grams of the dried sample, (see 4.3.1), into a tared porcelain dish. Ignite the sample with a small pin flame. The sulfur, once ignited, will burn evenly and clean unless organic matter is present. If organic matter is present, a film of melted asphalt or oily matter forms over the surface of the molten sulfur and shuts off the air so that the flame from the burning sulfur is extinguished. When a film of dark oily matter is first noticed, touch it lightly with the pin flame and it will usually break or char, allowing the sulfur to burn evenly. Toward the end of the burning, apply a very gentle heat to the dish and thus char the organic matter but keep the temperature well below red heat so that the organic matter is not ignited. When all the sulfur is burned off, as indicated by lack of sulfur dioxide odor, cool the dish in a desiccator and weigh. This weight, (A) represents the combined organic impurities and ash. Ignite the residue in the dish to burn off the organic impurities, cool in a desiccator and weigh. This weight (B) represents the ash, and the difference between the two (A-B) represents the organic impurities. Calculate the percentage of organic impurities and of ash as follows:

$$\text{Percent organic impurities} = \frac{(A - B) \times 100}{\text{weight of sample}}$$

$$\text{Percent ash} = \frac{B \times 100}{\text{weight of sample}}$$

4.3.3 Organic impurities retest.-If the percentage of organic impurities exceeds 0.1 percent as determined in 4.3.2, the percentage of sulfur contained in the residue shall be determined by the following procedure: weigh the same weight of sample as used in 4.3.2 into a tared porcelain dish. Ignite and burn the sample using the same procedure used in 4.3.2 until all the sulfur is burned off as indicated by lack of sulfur dioxide fumes. Prepare an oxidizing agent by melting 4 grams of potassium hydroxide and 0.5 grams of potassium nitrate in a large silver crucible with the addition of a little water. Cool the mixture and transfer the residue quantitatively from the tared porcelain dish to the large silver crucible. Heat gradually, stirring the mixtures frequently with a silver wire until all the organic matter is decomposed.

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Cool and dissolve the melt in approximately 300 milliliters of water. Make the solution just acid with hydrochloric acid. Heat to boiling and add 5 milliliters of a 10-percent barium chloride solution. Filter the solution through a tared Gooch crucible. Dry the precipitate in an oven for 1 hour, then in a muffle furnace to a dull red heat. Cool in a desiccator and weigh. The true percentage of organic impurities may be calculated as follows:

$$\text{Percent sulfur in residue} = \frac{A \times 100 \times 0.1373}{\text{weight of sample}}$$

$$\text{True percent organic impurities} = B - C$$

Where:

A = Weight of barium sulfate precipitate
B = Percent organic impurities as determined in 4.3.2.
C = Percent sulfur in residue as determined in this paragraph.

4.3.4 Sulfur, Major defect, Code No. 03001.-Determine the percent of sulfur as follows:

$$\text{Percent sulfur (dry basis)} = 100 - (\text{percent organic impurities} + \text{percent ash})$$

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.-Unless otherwise specified, the sulfur shall be packed in 5-ply multiwall paper bags containing approximately 112 pounds. The bags shall be constructed of 5-ply, 60-pound sheet (the outer sheet being highly calendered for water-resistance) 6/12 threads, pasted gussets, cushioned stitch, and sewn open mouth.

5.3 Bulk shipments.-When specified in the contract or order, bulk shipments of powder sulfur may be made in carload lots.

5.4 Marking.-Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, and quantity contained therein as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-The sulfur is intended for use in the manufacture of black powder.

6.4 Caution should be exercised in the storing of sulfur as it is a combustible material under certain conditions.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1375-A013

FSC 1375

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. *This sheet is provided for obtaining information on* the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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